

Theory

Explain React components

A React component is a reusable, independent piece of code that defines how a part of the user interface (UI) should appear and behave. Components help break a complex UI into smaller, manageable pieces.

For example, a webpage can be divided into components such as:

- Header
- Navigation Bar
- Sidebar
- Main Content
- Footer

Identify the differences between components and JavaScript functions.

React Component	JavaScript Function
Used to create UI elements.	Used to perform a specific task or calculation.
Returns JSX (HTML-like syntax).	Returns any JavaScript value (number, string, object, etc.).
Component names must begin with a capital letter.	Function names can start with lowercase or uppercase.
Can use React features like props, state, and hooks.	Cannot directly use React features unless it's a React component or custom hook.
Rendered using <code><ComponentName /></code> .	Called using <code>functionName()</code> .

Identify the types of components

Types of Components

React has two types of components:

1. Function Components
2. Class Components

Explain class component

A Function Component is a simple JavaScript function that accepts props as input and returns JSX.

Features

- Easy to write and understand.
- Uses React Hooks (useState, useEffect, etc.) for state and lifecycle.
- Preferred in modern React applications.
- Lightweight and faster.

Explain function component

A Class Component is an ES6 class that extends React.Component. It must include a render() method that returns JSX.

Features

- Supports state and lifecycle methods.
- More code than function components.
- Mostly used in older React applications.

Define component constructor

A constructor is a special method in a class component that is called automatically when the component is created.

Purpose

- Initialize the component's state.
- Bind event handler methods.

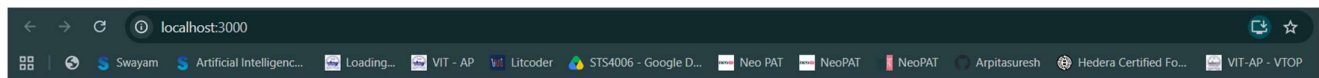
Define render() function

The render() method is a mandatory method in every class component.

Purpose

- Returns the JSX that should be displayed on the screen.
- Called whenever the component's state or props change.

Output of HANDS-ON



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Total: 284 Marks

Score: 94.67%